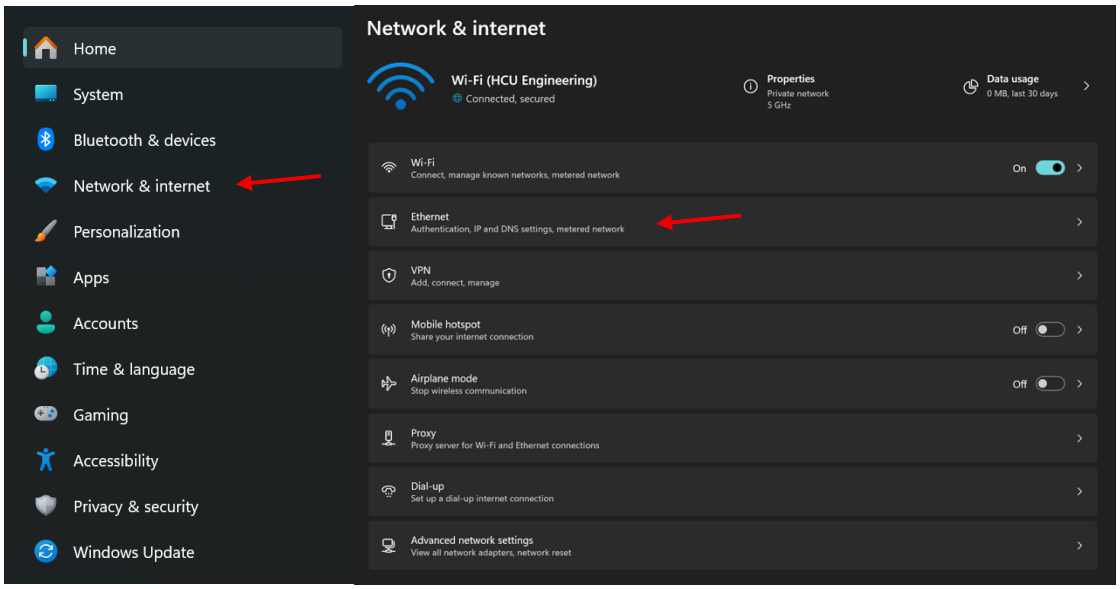
VNC Setup: Give Windows and Raspberry Pi Static IPs

Goal: Set fixed (static) IP addresses so you can reliably connect with VNC.

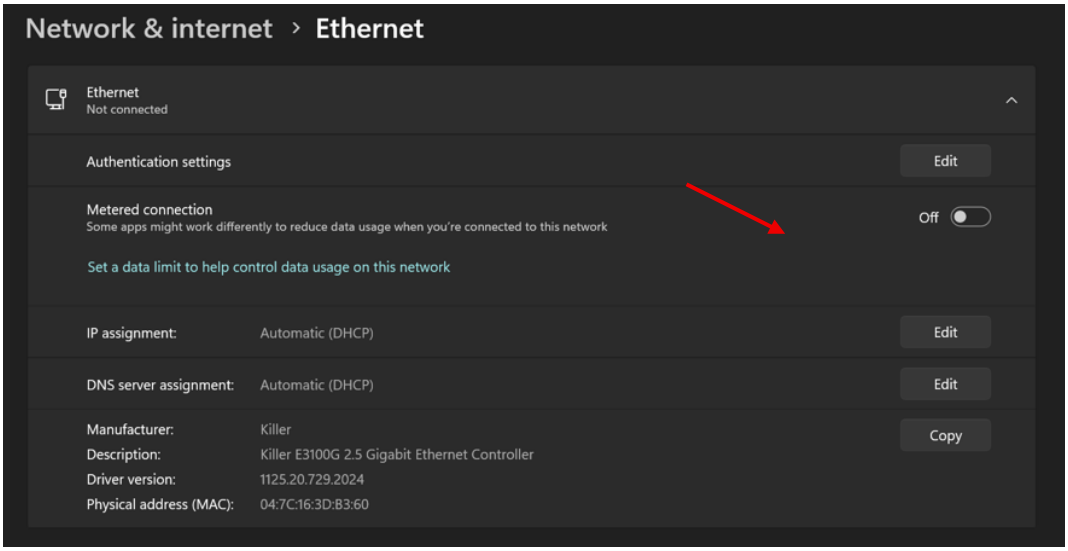
- Windows PC (Ethernet): **192.168.2.152 / 255.255.255.0**
- Raspberry Pi (wired): **192.168.2.151 / 255.255.255.0 (/24)**
- You'll then connect in VNC Viewer to **192.168.2.151**

Part 1 — Set a Static IP on Windows (Ethernet)

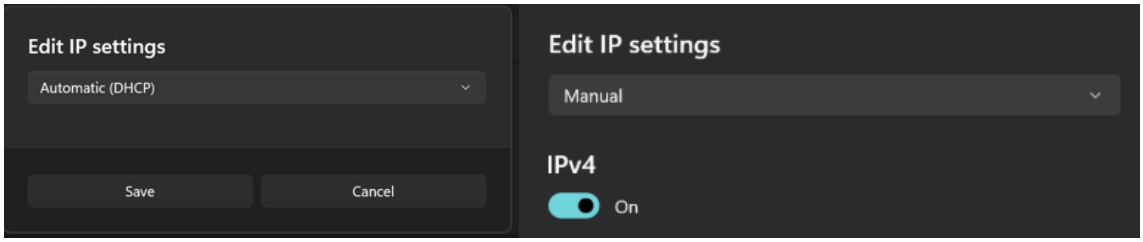
1. **Plug in Ethernet**
   - Make sure your Windows computer is connected to the Pi (directly or via a simple switch) with an Ethernet cable.
2. **Open Network Settings**
   - Click **Start** → **Settings** (gear icon).
   - Go to **Network & Internet** → **Ethernet** (make sure you pick the **wired**/Ethernet adapter you're using).



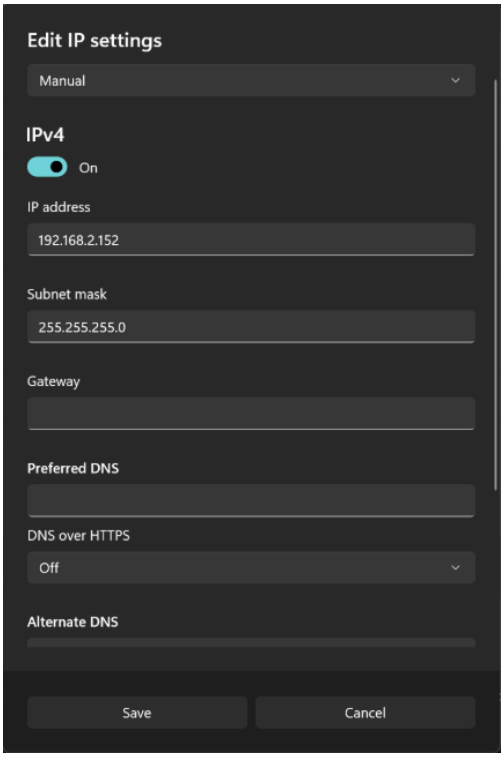
3. **Open IP settings**
   - Scroll to **IP assignment**. Click **Edit**.



4. **Switch to Manual**
   - In the dialog, change **Automatic (DHCP)** to **Manual**.
   - Toggle **IPv4** to **On** (if it isn't already).



5. **Enter the Static Details**
   - **IP address:** 192.168.2.152
   - **Subnet mask (if shown instead):** 255.255.255.0
   - **Gateway:** leave **blank**
   - **Preferred DNS:** leave

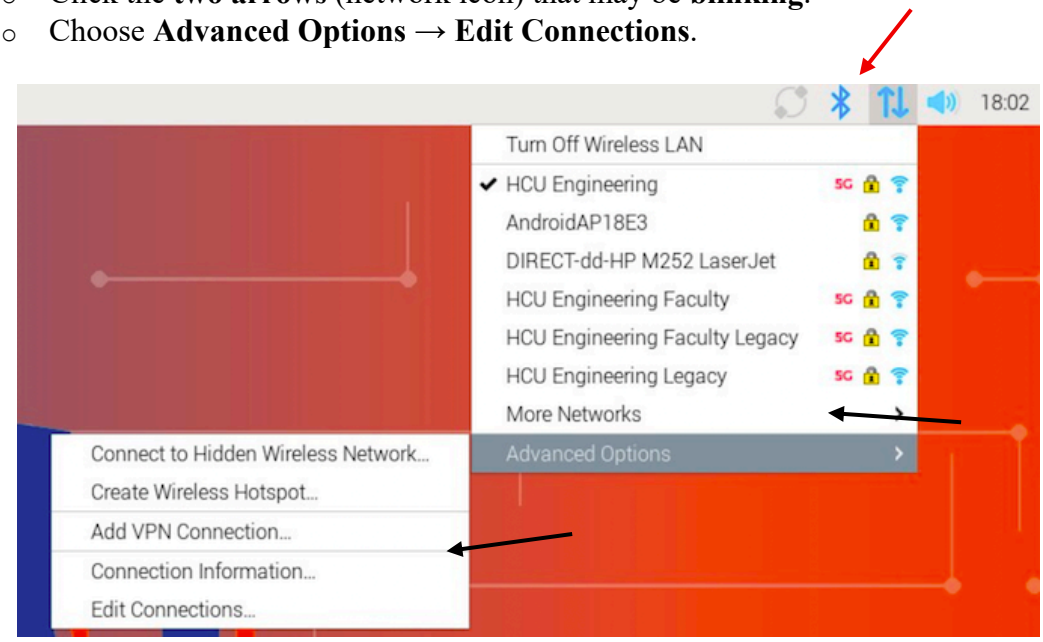


6. **Save**
   - Click **Save**.
   - You should briefly see the Ethernet status change and reconnect.
7. **Quick Check**
   - Press **Windows Key** → type **cmd** → open **Command Prompt**.
   - Type: ipconfig and press **Enter**.
   - Under your **Ethernet adapter**, confirm **IPv4 Address** shows 192.168.2.152 and **Subnet Mask** shows 255.255.255.0.

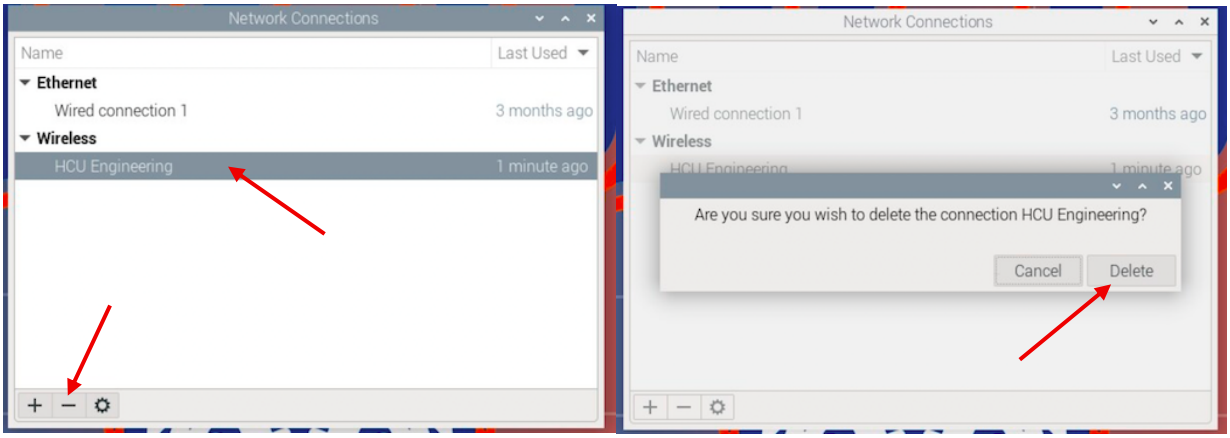
Part 2 — Set a Static IP on Raspberry Pi (wired)

You'll do this first using the Pi's current ".local" name, then switch to the static IP.

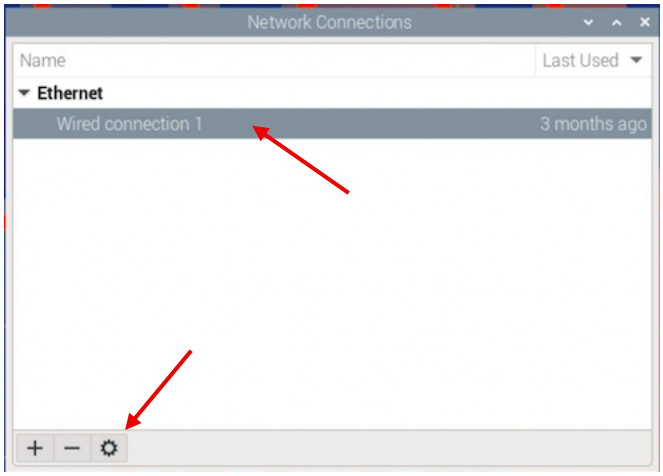
1. **Connect via VNC using the hostname**
   - Open **TigerVNC** or **RealVNC Viewer** on Windows.
   - In the address bar, enter **raspberrypi.local** and connect.
   - If prompted, accept security prompts and log in.
2. **Open Network Connections on the Pi**
   - On the Pi's desktop, look at the **top-right** icons.
   - Click the **two arrows** (network icon) that may be **blinking**.
   - Choose **Advanced Options → Edit Connections**.



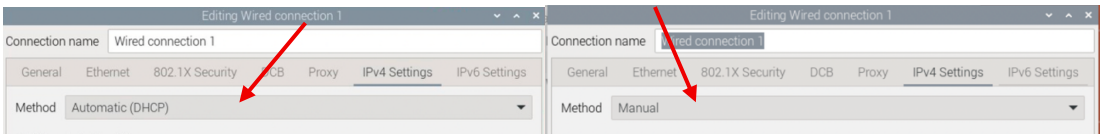
3. **Remove the HCU Engineering profile (if present)**
   - In the list, select **HCU Engineering**.
   - Click the **minus (–)** button to remove it.



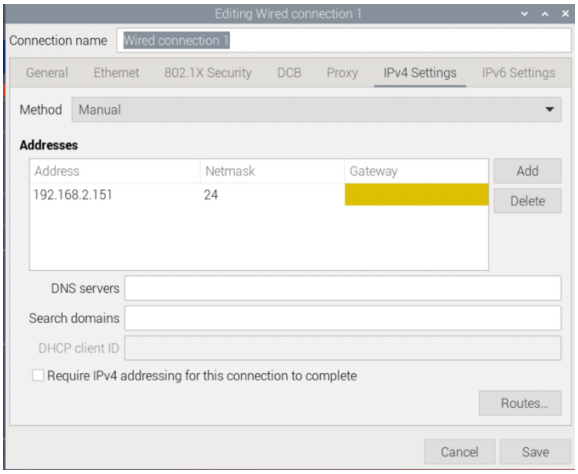
4. **Edit the Wired Profile**
   - Select **Wired connection** (or similar wired profile).
   - Click **Settings** (or **gear** icon).



5. **Change IPv4 to Manual**
   - Go to the **IPv4 Settings** tab.
   - Change **Method** from **Automatic (DHCP)** to **Manual**.



6. **Add the Static Address**
   - Click **Add** (under Addresses).
   - **Address:** 192.168.2.151
   - **Netmask:** 24 *(this equals 255.255.255.0)*
   - **Gateway:** leave **blank**
   - **DNS Servers:** leave **blank**
   - Click **Save**.



7. **Apply and Reconnect**
    - Close the network windows.
    - It may take a few seconds for the new IP to take effect.
8. **Close your current VNC session**
    - Close the VNC window you used for raspberrypi.local.
9. **Reconnect using the new static IP**
    - Open your VNC Viewer again.
    - In the address bar, enter **192.168.2.151** and connect.
    - Log in when prompted.

Part 3 — Verify & Troubleshoot (quick!)

If anything doesn't connect right away, try these in order:

1. **Ping test from Windows**
   - Open **Command Prompt** and type:

     ping 192.168.2.151

     - **Success:** You should see replies.
     - **If it fails:** Re-check that both devices' IPs and subnet are exactly as above, and that Ethernet is connected.
2. **Confirm Windows IP again**
   - ipconfig → look for **IPv4 Address** 192.168.2.152 and **Subnet Mask** 255.255.255.0.
3. **Confirm Pi IP locally (if you have keyboard/monitor on the Pi)**
   - Open a Terminal on the Pi and run:

     ip addr show eth0

     - Look for inet 192.168.2.151/24 on **eth0**.
4. **Make sure VNC is enabled on the Pi**
   - On the Pi: **Menu → Preferences → Raspberry Pi Configuration → Interfaces** tab → **VNC: Enable → OK**.
   - If you had to enable it, try reconnecting in VNC after a few seconds.
5. **Cable/port check**
   - Try another Ethernet cable or port if nothing else works.